# PENGOLAHAN CITRA DIGITAL

S1 RKA - 2026

# PRAPROSES CITRA DIGITAL

## Transformasi Fourier 2D

# TRANSFORMASI FOURIER 2D

Transformasi Fourier 2D membagi (*decompose*) sebuah citra digital menjadi komponen fungsi *sin* dan komponen fungsi *cos*.

Transformasi Fourier 2D mengubah sebuah citra dari domain spasial ke domain frekuensi.

Aplikasi transformasi Fourier: rekonstruksi citra digital, kompresi citra, perbaikan citra.

# TRANSFORMASI FOURIER 2D

Rumus Transformasi Fourier 2D:

$$F(k,l) = \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} f(i,j) e^{-i2\pi(\frac{ki}{N} + \frac{lj}{N})}$$

$$e^{ix} = \cos x + i \sin x$$

Hasil transformasi Fourier berupa bilangan kompleks.

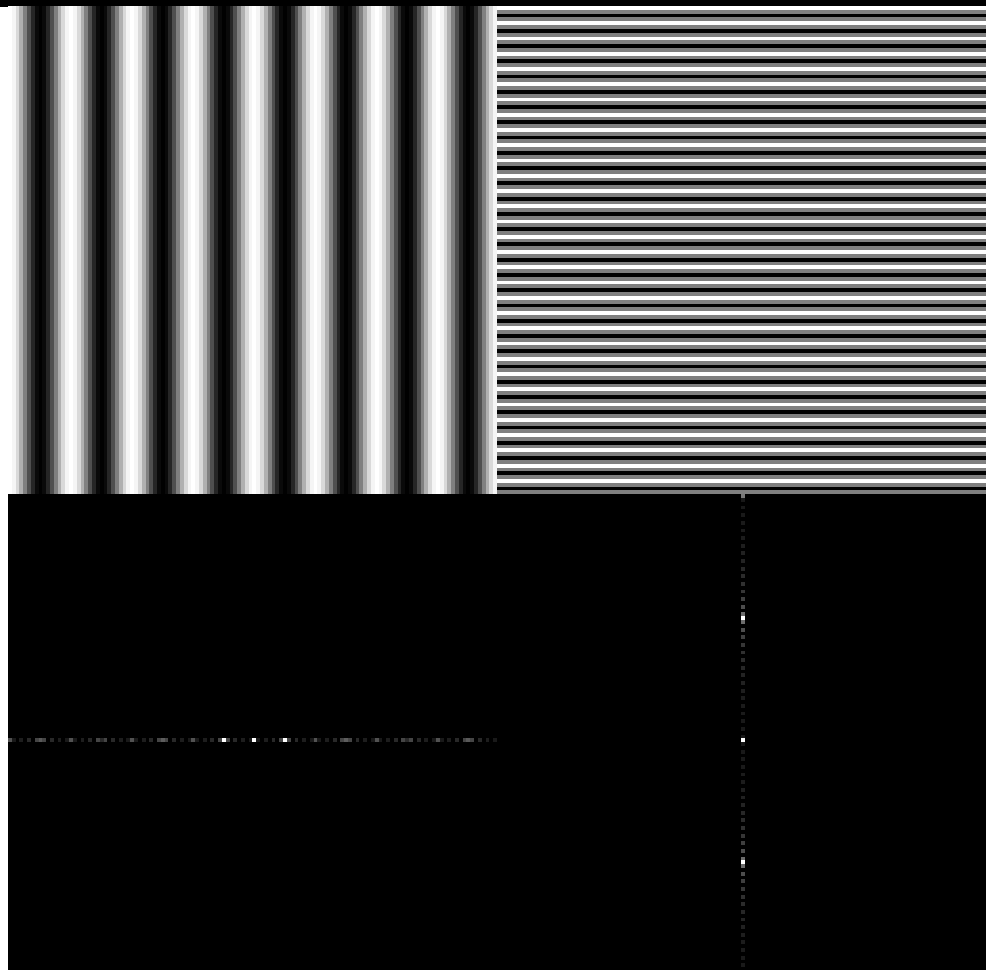
# SAMPLE IMAGES

Top Left image: a pure horizontal cosine of 8 cycles

Top Right image: a pure vertical cosine of 32 cycles

Bottom Left: the FT of the top left image

Bottom Right: the FT of the top right image



# SAMPLE IMAGES

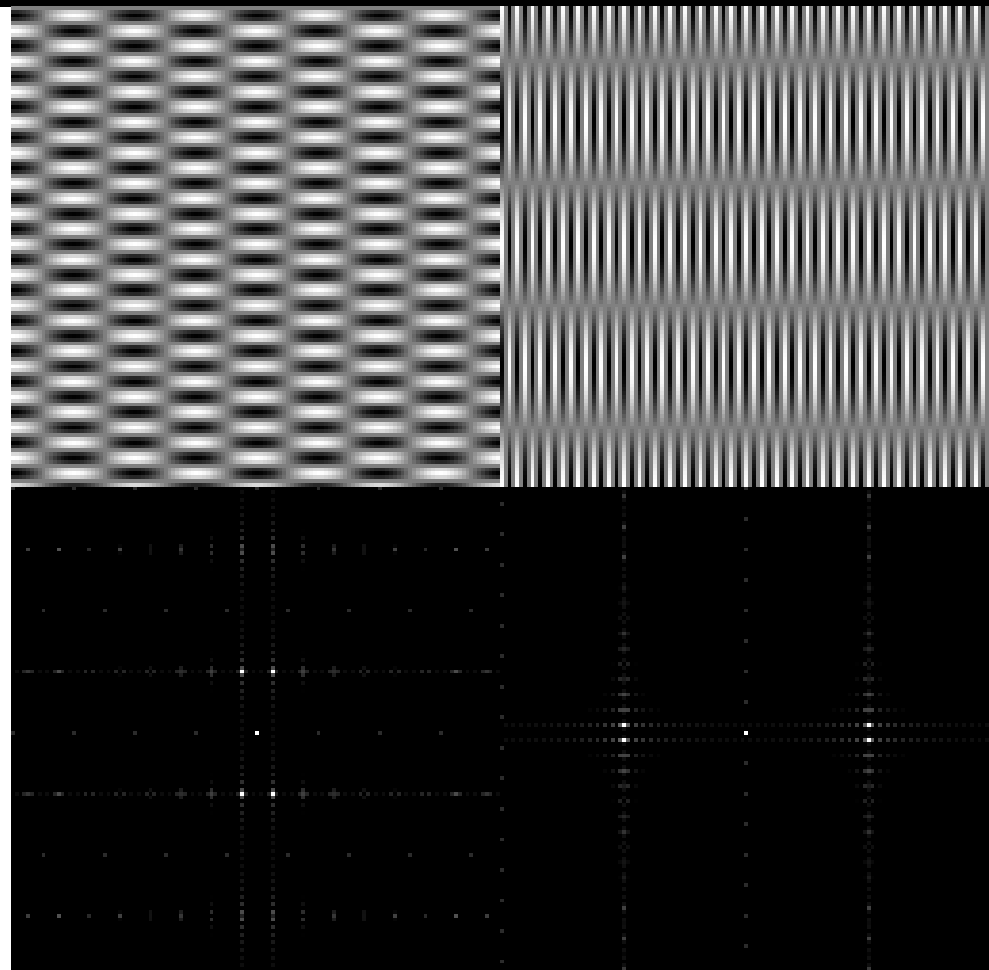
Top: images of 2D cosines with both horizontal and vertical components

Top left: has 4 cycles horizontally and 16 cycles vertically

Top right: has 32 cycles horizontally and 2 cycles vertically

Bottom Left: the FT of the top left image

Bottom Right: the FT of the top right image



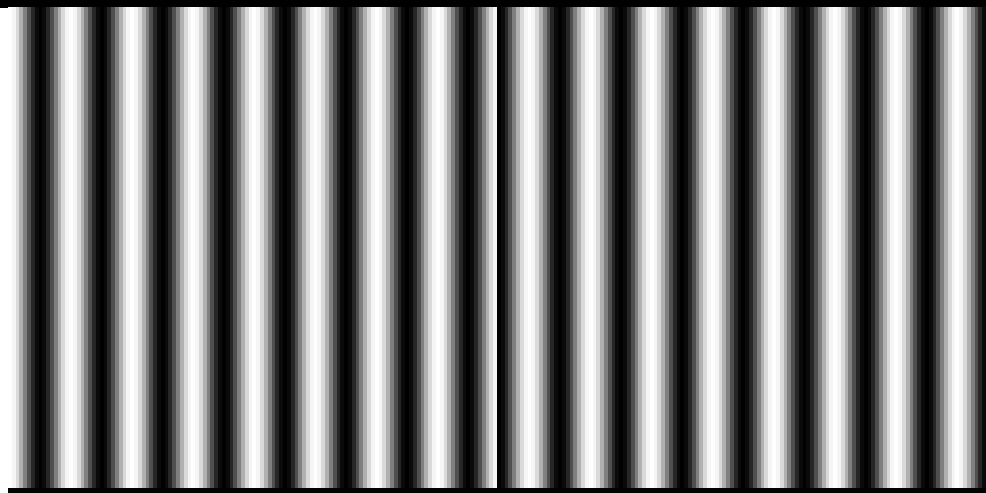
# MAGNITUDE AND PHASE

Top: images of horizontal cosines of 8 cycles

One image is shifted laterally from the other by 1/2 cycle (or by PI in phase)

Top right: has 32 cycles horizontally and 2 cycles vertically

Bottom Left: the FT of the top left image =
Bottom Right: the FT of the top right image



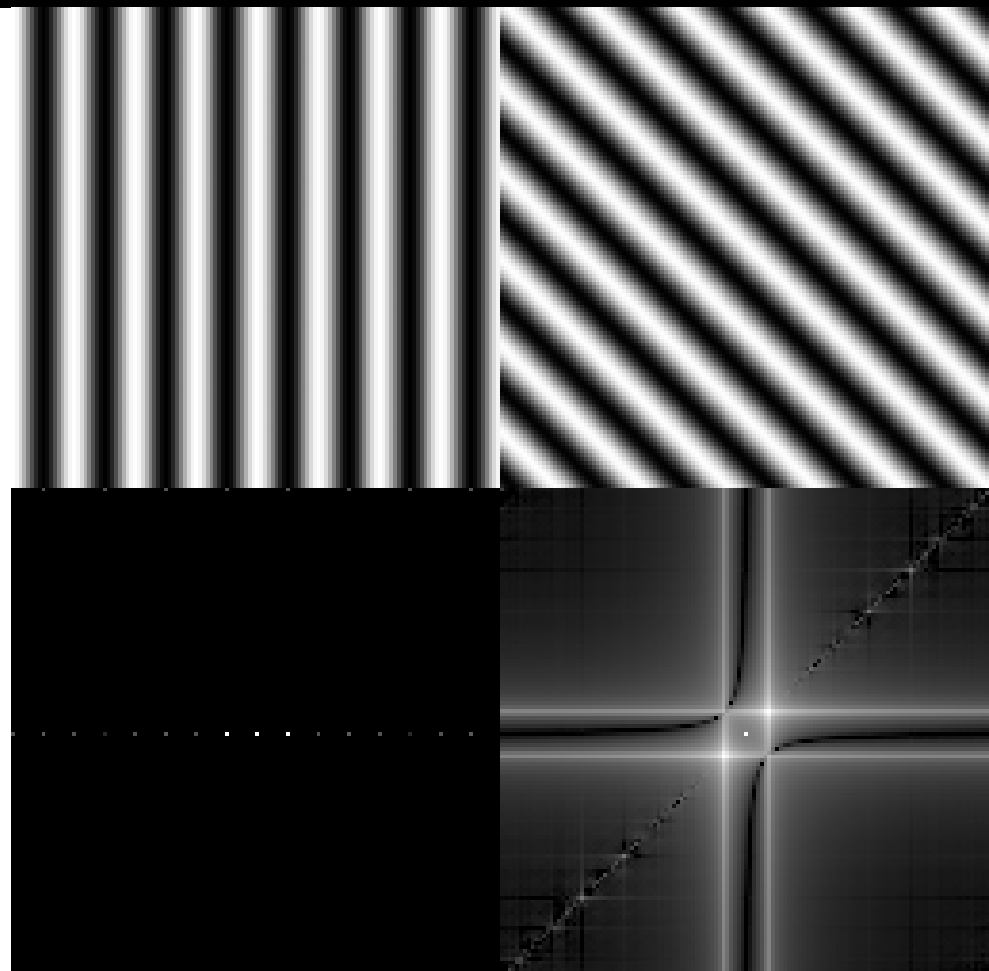
# ROTATION AND EDGE EFFECTS

Top Left: a pure horizontal cosine of 8 cycles

Top right: rotated cosine image

Bottom Left: the FT of the top left image

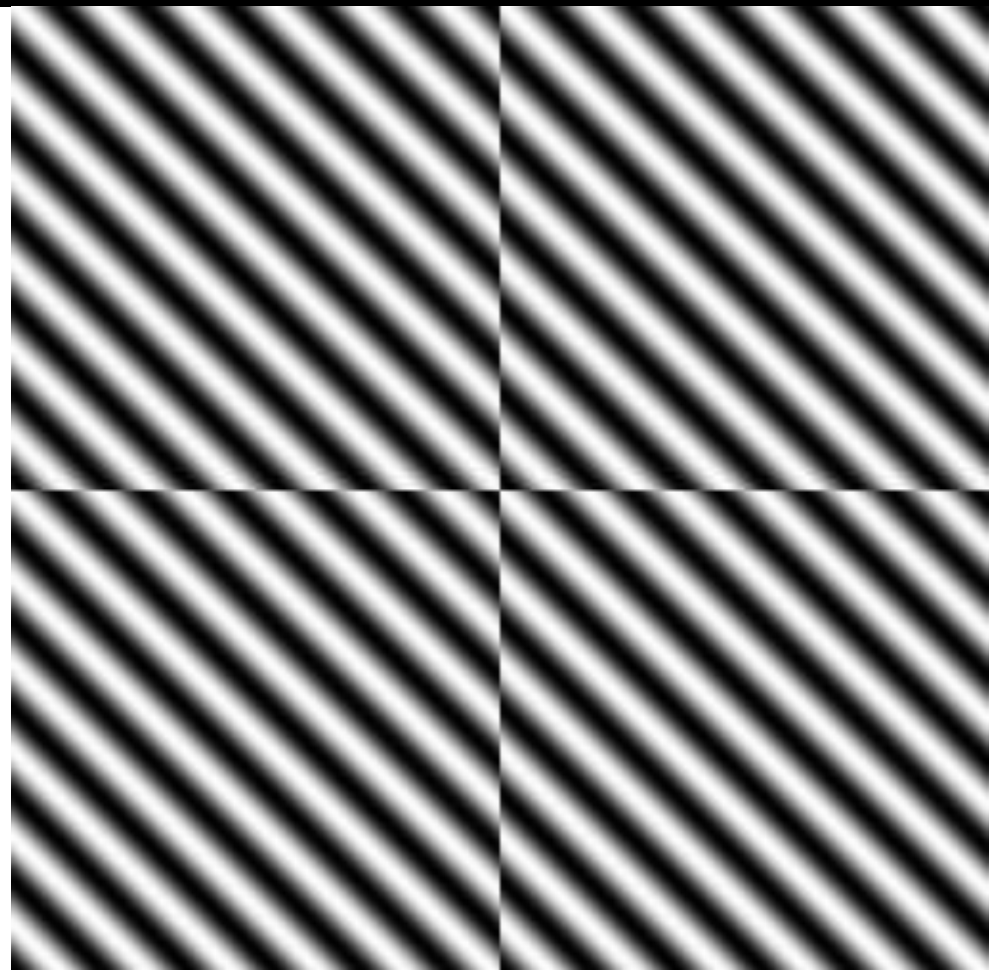
Bottom Right: the FT of the top right image



# ROTATION AND EDGE EFFECTS

The FT treats an image as if it were part of a periodically replicated array of identical images extending horizontally and vertically to infinity.

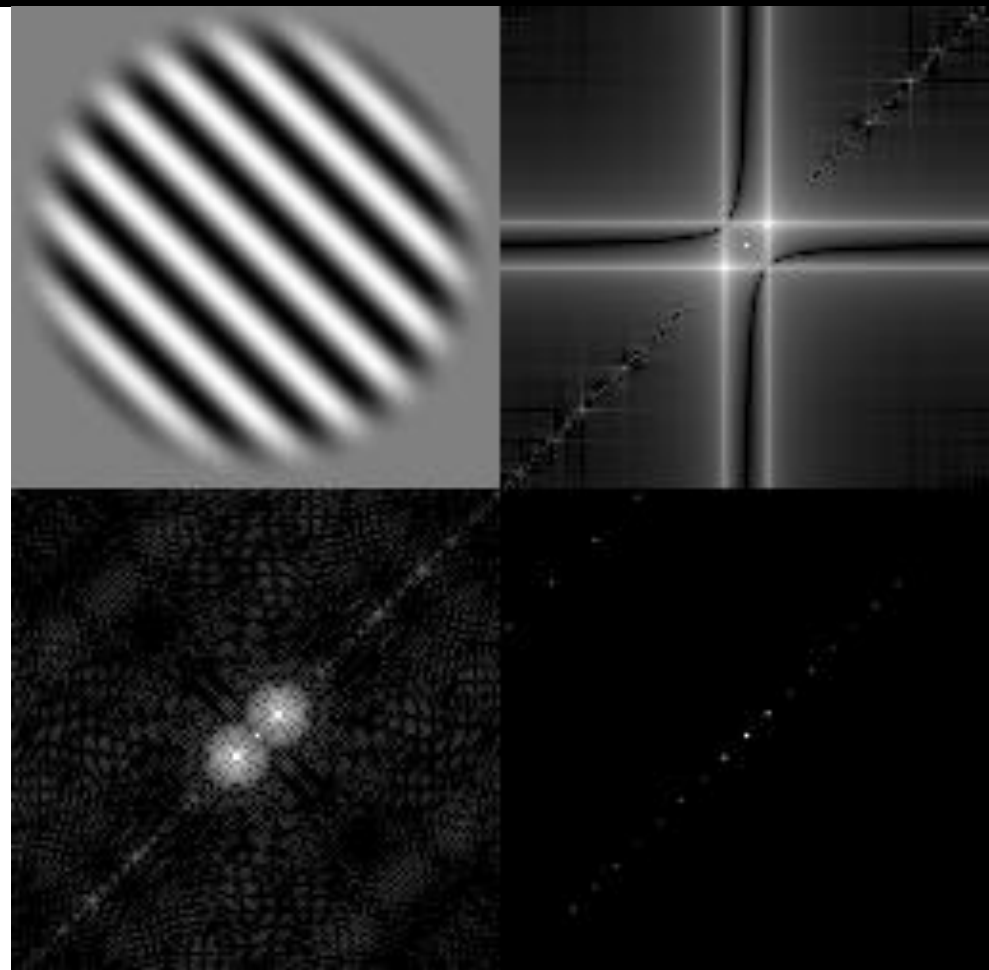
And there are strong edge effects between the neighbors of such a periodic array.



# ROTATION AND EDGE EFFECTS

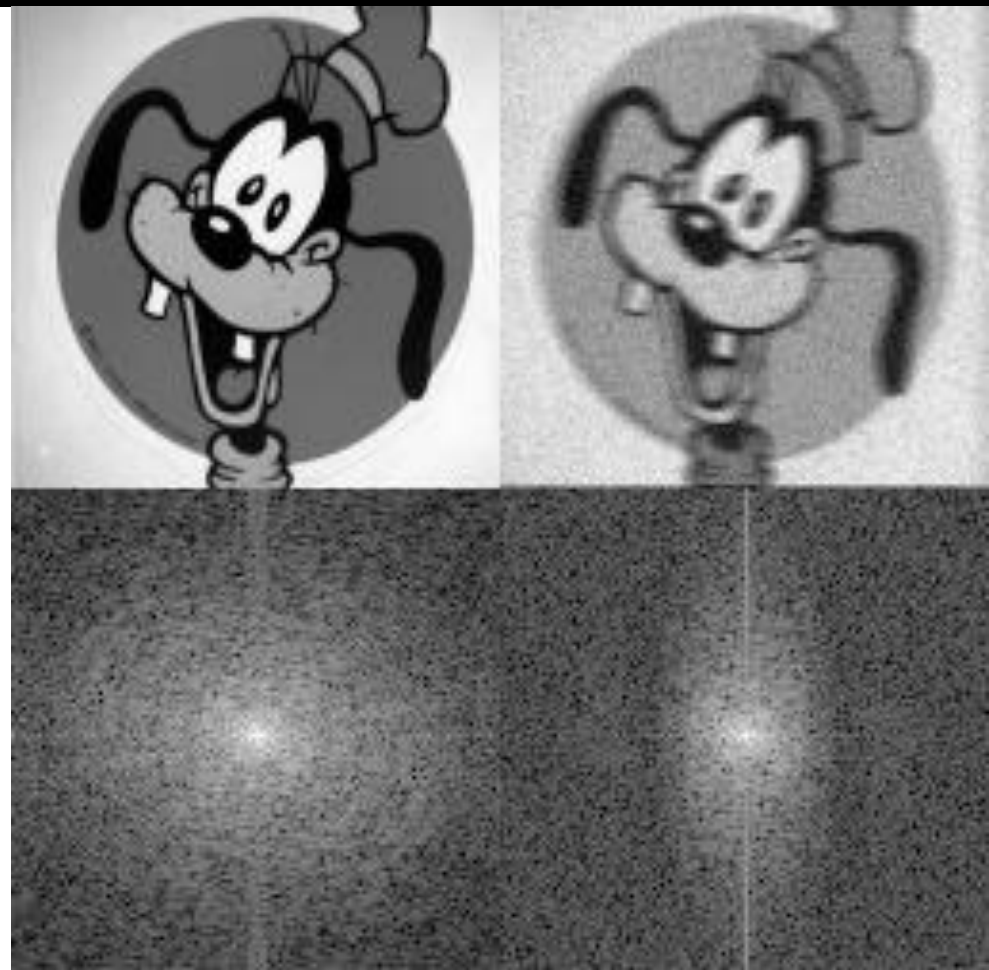
These edge effects can be significantly reduced by "windowing" the image with a function that slowly tapers off to a medium gray at the edge.

It can be seen that the windowed image FT is much closer to the true FT and eliminates many of the edge effects.



# SHARP AND CLEAR VS BLUR AND NOISY IMAGES

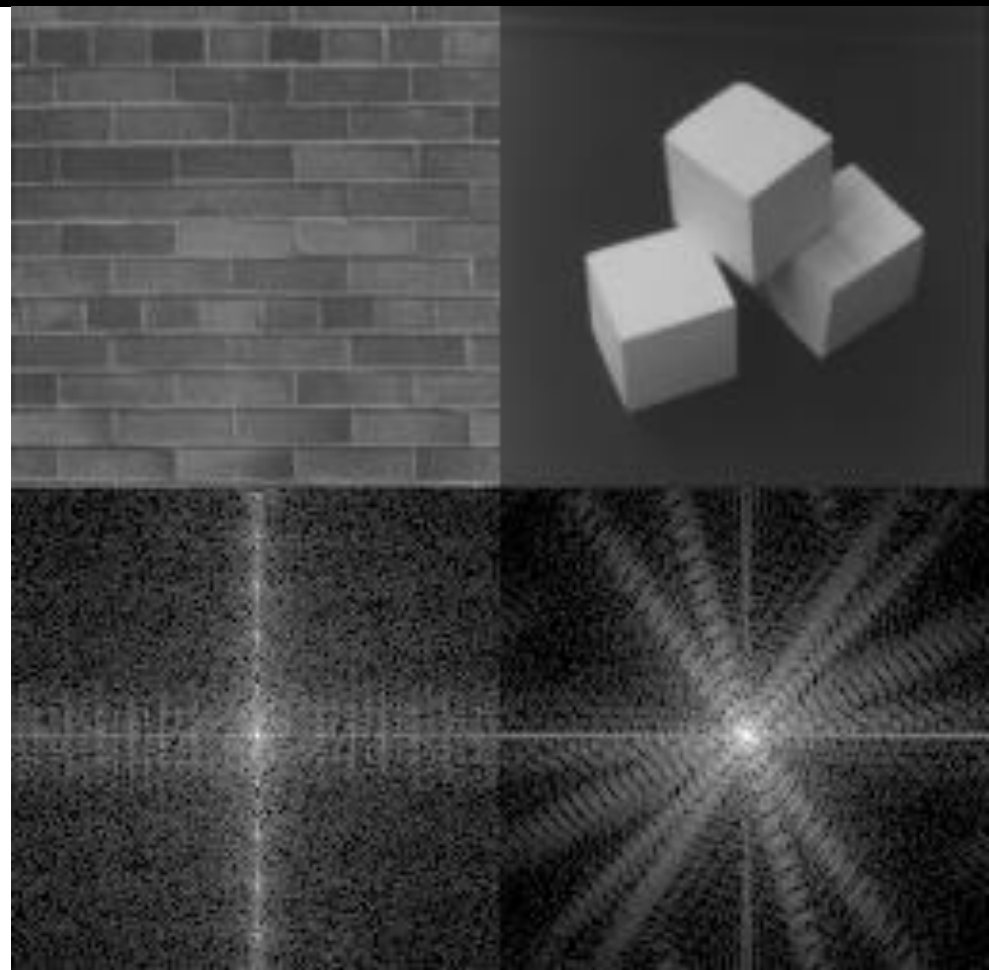
Both suffer from edge effects as evidenced by the strong vertical line through the center.



# EFFECTS OF EDGES IN IMAGES

Notice the strong periodic component, especially in the vertical direction for the bricks image. Horizontal components appear closer together in the FT.

In the blocks image, notice a bright line going to high frequencies perpendicular to the strong edges in the image.



# EFFECTS OF EDGES IN IMAGES

Notice that the letters have quite different FTs, especially at the lower frequencies.

The FTs also tend to have bright lines that are perpendicular to lines in the original letter.

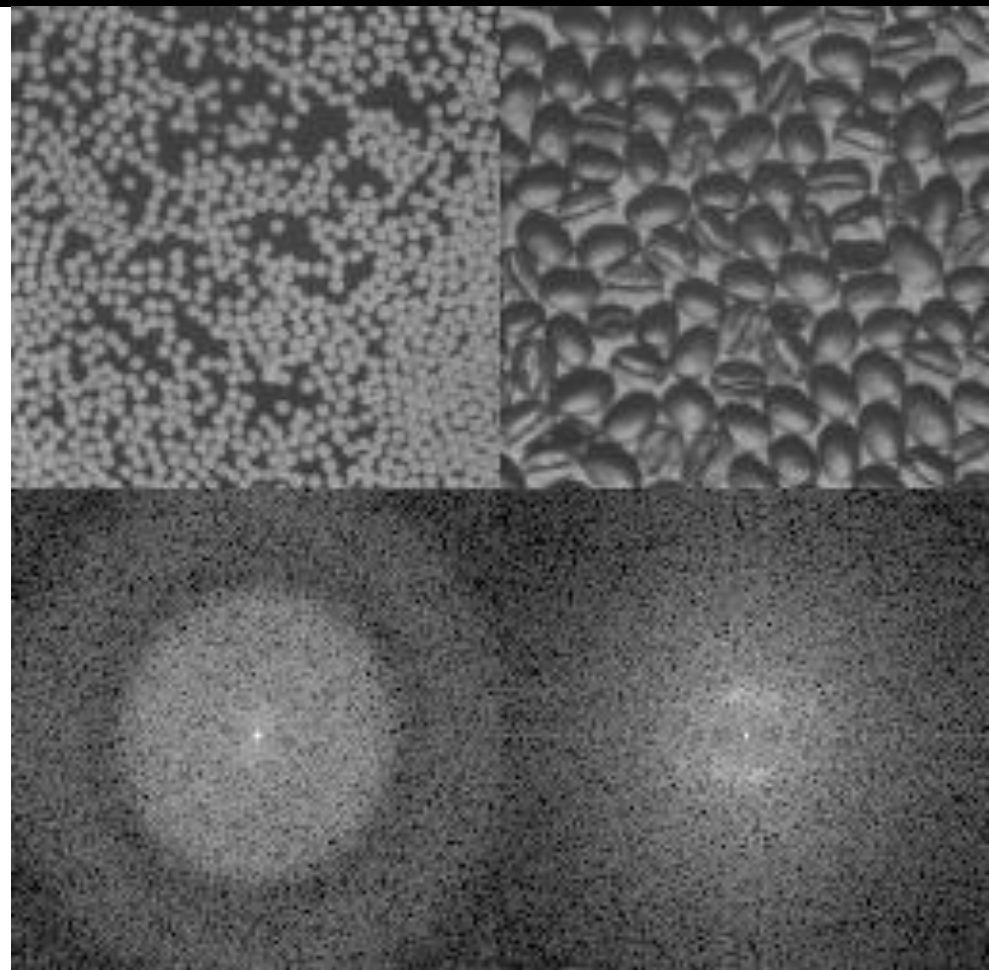
If the letter has circular segments, then so does the FT.



# SIMILAR OBJECTS

The concentric ring structure in the FT of the white pellets image is due to each individual pellet.

The coffee beans have less symmetry and are more variably colored so they do not show the same ring structure.

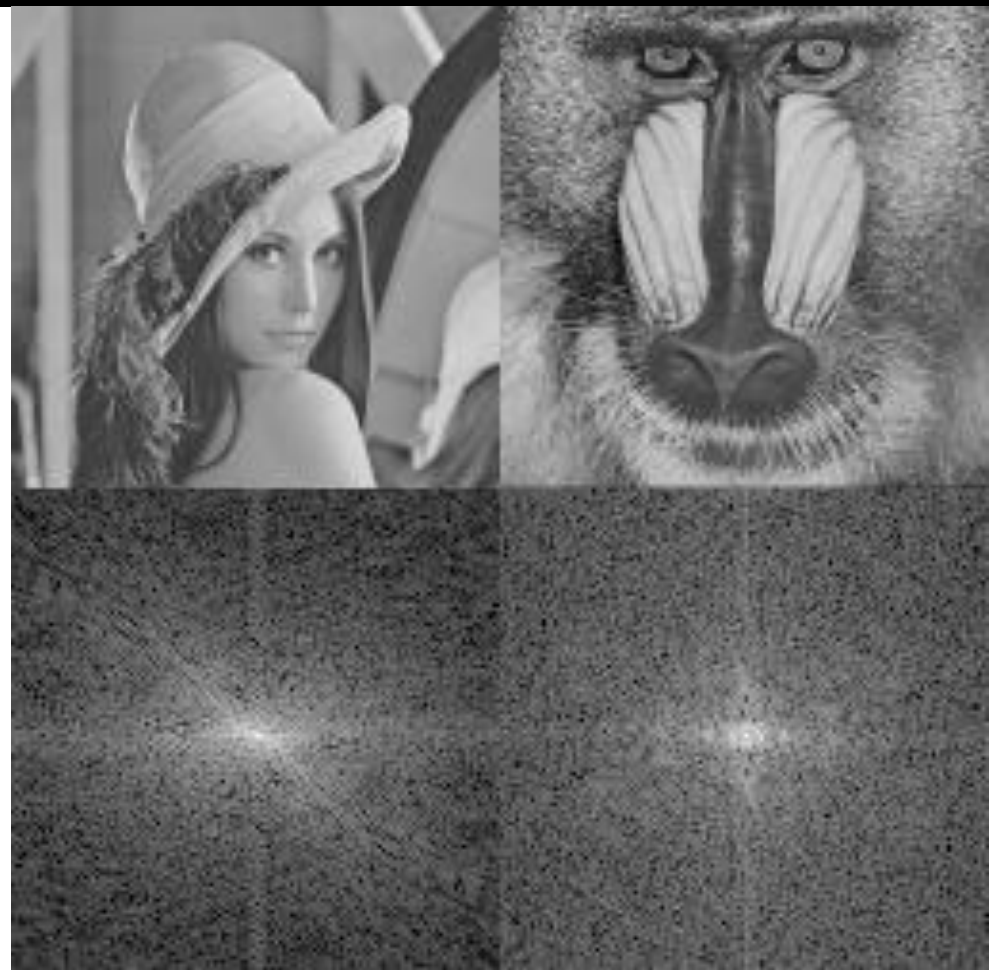


# GENERAL IMAGES

Notice there is very little structure.

There is a top left to bottom right slanting line in the girl image FT, due to the edge between her hat and her hair.

The mandril image appears to have more high frequency power, probably due to the hair.



# PRAPROSES CITRA DIGITAL

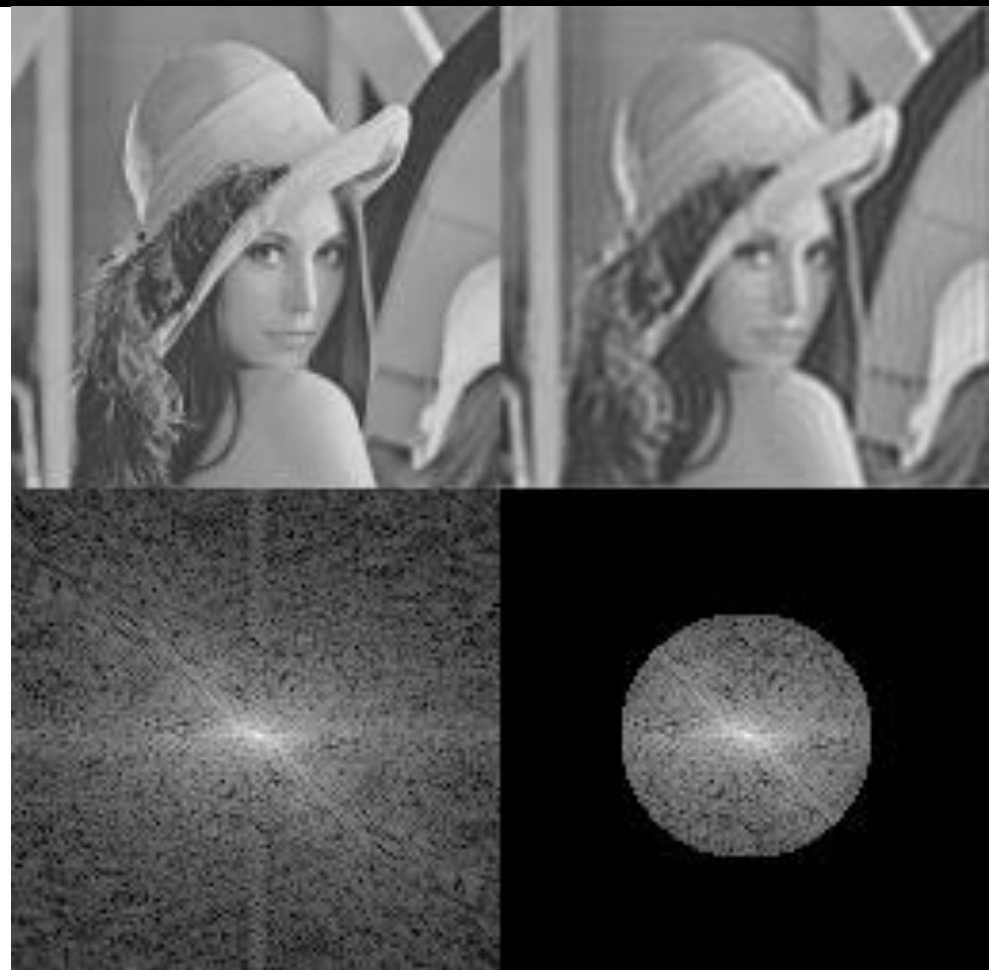
Pemfilteran citra dengan *lowpass* dan *highpass* dalam domain frekuensi

# LOWPASS FILTERS

The left side of the image we have seen before.

Bottom right: the high frequencies are cut off by the "ideal" lowpass filter.

Top right: reconstructed image using inverse Fourier transformation, which is blurrier due to the loss of high frequencies.



# HIGHPASS FILTERS

Bottom right: the center bright spot, which accounts for most of the power in the image, is clearly gone.

Top right: reconstructed image using inverse Fourier transformation, which looks totally black, but in fact is not.

The highpass filter has preserved the image information where there are very rapid changes in gray level.



# SHARPENING USING HIGH EMPHASIS FILTERS

**Bottom right:** Fourier coefficients of frequency-distance 0 from the origin will be multiplied by 0.5. The multiplier will be interpolated between 0.5 and 4.0 from the zero frequency to frequency-distance 96. From then outward, the multiplier will be 4.0.

Notice the relative brightness at high frequencies.

**Top right:** reconstructed image using inverse Fourier transformation, which is sharper.



# AKTIVITAS MINGGU 4

Pelajari image processing toolbox/library

Pelajari/Implementasikan materi image enhancement pada domain frekuensi dengan image processing toolbox/library tsb

# TERIMA KASIH